

MATH1013 — CHEAT SHEET

EULER'S METHOD — SETUP

ODE $\frac{dx}{dt} = F(x, t)$. Euler approx: $x_n = x_{n-1} + \Delta t \cdot F(x_{n-1}, t_{n-1})$, where $t_n = n\Delta t$. This gives a 1st-order DE. Solve it the usual way. Then check it recovers the exact ODE solution as $\Delta t \rightarrow 0$.
Limit tools: $a^n = (e^{\Delta t})^n / \Delta t = e^t \cdot (1 - \Delta t)^n \rightarrow e^{-t}$ (Taylor). L'Hôpital for prefactor.

EXAM Q4 — FULL EULER WALKTHROUGH

ODE: $\frac{dx}{dt} = -x + 2e^t$, $x(0) = 0$. (a) Verify exact soln $x = e^t - e^{-t}$. (b) Show Euler gives $x_n = (1 - \Delta t)x_{n-1} + \frac{2\Delta t}{e} a^n$ where $a = e^{\Delta t}$. (c) Solve, show $x_n = h(\Delta t)(a^n - (1 - \Delta t)^n)$, find h . (d) Take $\Delta t \rightarrow 0$, recover exact soln.

(a) **Verify by substitution:**
Compute LHS: $\frac{d}{dt}(e^t - e^{-t}) = e^t + e^{-t}$
Compute RHS using proposed x : $-x + 2e^t = -(e^t - e^{-t}) + 2e^t = -e^t + e^{-t} + 2e^t = e^t + e^{-t}$
LHS = RHS ✓. Check IC: $x(0) = e^0 - e^0 = 1 - 1 = 0$ ✓
(b) **Derive Euler DE:**
Sub $F(x, t) = -x + 2e^t$ into Euler formula:
 $x_n = x_{n-1} + \Delta t(-x_{n-1} + 2e^{\Delta t(n-1)})$
 $= x_{n-1} - \Delta t \cdot x_{n-1} + 2\Delta t \cdot e^{\Delta t(n-1)}$
 $= (1 - \Delta t)x_{n-1} + 2\Delta t \cdot a^{n-1}$ where $a = e^{\Delta t}$
 $= (1 - \Delta t)x_{n-1} + 2\Delta t \cdot \frac{a^n}{a}$
 $x_n = (1 - \Delta t)x_{n-1} + \frac{2\Delta t}{a} \cdot a^n$ ✓
(c) **Solve the DE:**
This is 1st-order: $r = (1 - \Delta t)$, forcing $= \frac{2\Delta t}{a} \cdot a^n$. Since $a = e^{\Delta t} \neq (1 - \Delta t)$, no resonance.
Trial: $A \cdot a^n$.
Sub: $Aa^n = (1 - \Delta t)Aa^{n-1} + \frac{2\Delta t}{a} a^n$
Divide through by a^{n-1} : $Aa = (1 - \Delta t)A + 2\Delta t$
 $A(a - 1 + \Delta t) = 2\Delta t \Rightarrow A = \frac{2\Delta t}{a - 1 + \Delta t}$
General soln: $x_n = C(1 - \Delta t)^{n-1} + \frac{2\Delta t}{a - 1 + \Delta t} \cdot a^n$
Sub $x_0 = 0$: $C + \frac{2\Delta t}{a - 1 + \Delta t} = 0 \Rightarrow C = -\frac{2\Delta t}{a - 1 + \Delta t}$
 $x_n = \frac{2\Delta t}{a - 1 + \Delta t} (a^n - (1 - \Delta t)^n)$
 $h(\Delta t)$
(d) **Take $\Delta t \rightarrow 0$:**
Term 1: $n = t/\Delta t$, so $a^n = (e^{\Delta t})^{t/\Delta t} = e^t$
Term 2: $(1 - \Delta t)^n = \exp(n \ln(1 - \Delta t)) = \exp(\frac{t}{\Delta t} \ln(1 - \Delta t))$
Taylor: $\ln(1 - \Delta t) = -\Delta t - \frac{(\Delta t)^2}{2} - \dots$
So $\frac{1}{\Delta t} \ln(1 - \Delta t) = \frac{1}{\Delta t} \left(-\Delta t - \frac{(\Delta t)^2}{2} - \dots \right) = -1 - \frac{\Delta t}{2} - \dots \rightarrow -t$
 $(1 - \Delta t)^n \rightarrow e^{-t}$
Prefactor $h(\Delta t) = \frac{2\Delta t}{a - 1 + \Delta t}$. As $\Delta t \rightarrow 0$, this is $\frac{2}{e}$. Apply L'Hôpital:
Differentiate top w.r.t. Δt : 2. Bottom: $\frac{d}{d\Delta t}(e^{\Delta t} - 1) = e^{\Delta t} \rightarrow 1$ as $\Delta t \rightarrow 0$.
 $h(\Delta t) \rightarrow \frac{2}{e} = 1$
Therefore $x_n \rightarrow 1 \cdot (e^t - e^{-t}) = e^t - e^{-t}$ ✓ — matches exact soln.

EXTRA Q2 — EULER FOR $DX/Dt = 1 - X$

Euler gives $x_n = x_{n-1} + \Delta t(1 - x_{n-1})$. (a) Solve, $x_0 = 0$, behaviour as $n \rightarrow \infty$. Sketch $\Delta t = 0.5$ and $\Delta t = 1.5$. (b) $\Delta t = 0.1$: steps to within 1% of limit.

(a) $x_n = (1 - \Delta t)x_{n-1} + \Delta t$. Trial const: $A = (1 - \Delta t)A + \Delta t \Rightarrow \Delta t \cdot A = \Delta t \Rightarrow A = 1$.
General: $C(1 - \Delta t)^n + 1$. Sub $x_0 = 0$: $C = -1$.
 $x_n = 1 - (1 - \Delta t)^n$. As $n \rightarrow \infty$; $x_n \rightarrow 1$ (provided $|1 - \Delta t| < 1$, i.e. $\Delta t < 2$).
 $\Delta t = 0.5$: $r = 0.5 > 0 \rightarrow$ monotone convergence to 1. $\Delta t = 1.5$: $r = -0.5 < 0 \rightarrow$ alternating convergence to 1. $\Delta t = 2$: $r = -1$, oscillates forever. $\Delta t > 2$: diverges.
(b) Within 1% means $|x_n - 1| < 0.01$, i.e. $(1 - \Delta t)^n < 0.01$. With $\Delta t = 0.1$:
 $(0.9)^n < 0.01 \Rightarrow n \ln 0.9 < \ln 0.01 \Rightarrow n > \frac{\ln 0.01}{\ln 0.9} \approx \frac{-4.605}{-0.105} = 43.7$
44 steps. Exact ODE: $x = 1 - e^{-t}$, $e^{-t} = 0.01 \Rightarrow t = 4.605$, i.e. **46 steps. Euler converges faster (since $x_n > x(n\Delta t)$).**

EXTRA Q3 — 2ND-ORDER FIXED PT WITH PARAMETER K

$x_n = 2 + \frac{1}{k}(x_{n-1}^2 - x_{n-2}^2)$, $k > 1$. (a) Find fixed pt. (b) Stable? (c) How does behaviour depend on k ?
(a) At fixed pt $x_n = x_{n-1} = x_{n-2} = x_f$:
 $x_f = 2 + \frac{1}{k}(x_f^2 - x_f^2) = 2$
 $x_f = 2$
(b) Set $x_n = 2 + q_n$ (small q). Then $x_{n-1}^2 - x_{n-2}^2 = (2 + q_{n-1})^2 - (2 + q_{n-2})^2$.
 $= (4 + 4q_{n-1} + q_{n-1}^2) - (4 + 4q_{n-2} + q_{n-2}^2) \approx 4q_{n-1} - 4q_{n-2}$ (drop q^2 terms)
So $q_n = \frac{4}{k}q_{n-1} - \frac{4}{k}q_{n-2}$
Char. eq: $\lambda^2 - \frac{4}{k}\lambda + \frac{4}{k} = 0$
 $\lambda = \frac{\frac{4}{k} \pm \sqrt{(\frac{4}{k})^2 - 4 \cdot \frac{4}{k}}}{2} = \frac{1}{k}(1 \pm \sqrt{1 - k})$
Since $k > 1$: $\sqrt{1 - k} = i\sqrt{k - 1} \rightarrow$ complex roots.
 $\rho^2 = |\lambda|^2 = (\frac{1}{k})^2(1^2 + (k - 1)) = \frac{1}{k} \cdot k = \frac{1}{k}$
So $\rho = \sqrt{\frac{1}{k}} = \frac{1}{\sqrt{k}}$. Stable iff $\rho < 1 \Rightarrow \frac{1}{\sqrt{k}} < 1 \Rightarrow k > 4$.
Stable if $k > 4$; unstable if $k \leq 4$
(c) For any $k > 1$: complex roots $\rightarrow$ oscillatory behaviour near $x_f = 2$.
 $k > 4$: decaying oscillations (spiral in). (V)

SAMPLE Q4 — 2ND-ORDER FIXED PT (CUBIC)

$x_n = 2 + \frac{1}{k}(x_{n-1}^3 - x_{n-2}^3)$. (a) Find fixed pt. (b) Is it stable? (c) Describe behaviour.
(a) At fixed pt: $x_f = 1 + \frac{1}{k}(x_f^3 - x_f^3) = 1$.
 $x_f = 1$
(b) Set $x_n = 1 + q_n$. Expand: $x^3 = (1 + q)^3 = 1 + 3q + 3q^2 + q^3 \approx 1 + 3q$ for small q .
 $x_{n-1}^3 - x_{n-2}^3 \approx (1 + 3q_{n-1}) - (1 + 3q_{n-2}) = 3q_{n-1} - 3q_{n-2}$
So $q_n = \frac{3}{k}(3q_{n-1} - 3q_{n-2}) = \frac{3}{k}q_{n-1} - \frac{3}{k}q_{n-2}$
Char. eq: $\lambda^2 - \frac{3}{k}\lambda + \frac{3}{k} = 0 \Rightarrow 2\lambda - \lambda - 1 = 0$
 $\lambda = \frac{\frac{3}{k} \pm \sqrt{(\frac{3}{k})^2 - 4 \cdot \frac{3}{k}}}{2} = \frac{1 \pm \sqrt{1 - k}}{2}$
 $\rho = |\lambda| = \sqrt{(\frac{1}{2})^2 + (\frac{\sqrt{k-1}}{2})^2} = \sqrt{\frac{1+k}{4}} = \frac{\sqrt{1+k}}{2} \approx \frac{\sqrt{2}}{2} \approx 0.707 < 1$
Fixed pt is stable.
(c) Complex roots with $\rho < 1 \rightarrow$ oscillatory decay towards $x_f = 1$ as $n \rightarrow \infty$.

PARTICULAR SOLUTION TABLE

$f(n)$	Trial $x^{(0)}$	Resonance (if trial = homog)
k	A	An if $r = 1$
kn	$An + B$	multiply trial by n
kn^2	$An^2 + Bn + D$	
ka^n , $a \neq r$	Aa^n	if $a = r$: try $An \cdot r^n$
$k \cdot r^n$	$An \cdot r^n$	resonance — see EG above
$k(-1)^n$	$A(-1)^n$	if $r = -1$: $An(-1)^n$

Combo forcing: sum of trials. Sub whole thing back in, equate coefficients separately.

HOW TO FIND C_1, C_2 FROM ICS

After writing general soln $c_1\lambda_1^n + c_2\lambda_2^n + x^{(0)}$:
Sub $n = 0$: $c_1 + c_2 + x^{(0)}|_{n=0} = x_0 \rightarrow$ equation 1.
Sub $n = 1$: $c_1\lambda_1 + c_2\lambda_2 + x^{(0)}|_{n=1} = x_1 \rightarrow$ equation 2.
Solve the 2x2 system for c_1, c_2 .

QUICK RECIPE — ALL QUESTION TYPES

1st-order DE: ① r = coeff of x_{n-1} ② pick trial from table ③ sub, equate coeffs $\rightarrow$ A, B ④ general = $Cx^n + x^{(0)}$, sub $x_0 \rightarrow C$
2nd-order DE: ① char. eq: $\lambda^2 - p\lambda - q = 0$ ② roots $\rightarrow$ homog ③ trial $\rightarrow$ particular ④ sub $x_0, x_1 \rightarrow 2$ eqs $\rightarrow c_1, c_2$
Fixed pts (1st order): ① solve $x = f(x)$ ② compute $f'(x_f)$ ③ $|f'| < 1 \rightarrow$ stable; $\text{sign} \rightarrow$ alt or not
2-cycle: ① confirm $|f'(x_f)| > 1$ ② compute $g = f \circ f$ ③ solve $g(x) = x$, remove x_f ④ $g'(a) = f'(a) \cdot f'(b)$: stable if $|g'| < 1$
2nd-order fixed pt: ① solve $x_f = f(x_f, x_f)$ ② set $q_n = x_n - x_f$, Taylor, drop q^2 ③ char. eq. $\rightarrow \lambda \rightarrow$ compute $\rho = |\lambda|$ ④ stable iff $\rho < 1$
Euler: ① sub F to rearrange to $r \cdot x_{n-1} + K a^n$ ② trial $Aa^n \rightarrow h(\Delta t)$ ③ $C = -h$ ④ limit: $a^n \rightarrow e^t$, $(1 - \Delta t)^n \rightarrow e^{-t}$, $h \rightarrow 1$ (L'Hôpital)
Matrix: ① $M = \begin{pmatrix} p & q \\ r & 0 \end{pmatrix}$ ② eigenvalues ③ eigenvectors ④ gen soln $\sum c_i e_i \lambda_i^n$
Applied model: ① write $x_n = r \cdot x_{n-1} + \text{added}$ ② solve ③ steady state = particular soln ④ reach level L : $n = \frac{\ln L}{\ln r}$, round up

BEHAVIOUR AS $N \rightarrow \infty$ — FULL ANALYSIS

In general soln: each term λ_i^n grows if $|\lambda_i| > 1$, decays if $|\lambda_i| < 1$, constant if $|\lambda_i| = 1$.
The dominant term is the one with **largest** $|\lambda_i|$.
If all $|\lambda_i| < 1$ and there's polynomial forcing $An^k + \dots$, that polynomial dominates: $x_n/n^k \rightarrow A$.
Sign of λ : $\lambda > 0 \rightarrow$ non-alternating. $\lambda < 0 \rightarrow$ alternating (sign flips each step).
Complex λ : oscillatory (neither purely alt nor non-alt), stable iff $\rho < 1$.

EXAM Q2c example: $x_n = \lambda_1^n + \lambda_2^n + 2n - 3$, $|\lambda_1| < 1$, $|\lambda_2| < 1$. Both λ^n terms $\rightarrow 0$. Forcing $2n - 3$ grows. As $n \rightarrow \infty$: $x_n/n \rightarrow 2$.
Extra Q4b example: $x_n = \frac{11}{5} \cdot 4^n + \frac{9}{5}(-1)^n - 1$, $|\lambda_1| = 4 > 1$. Dominant term: 4^n . **Grows without bound (non-alternating since $4 > 0$).**
Complex roots example: $\rho = \frac{1}{\sqrt{2}}$, $\rho^n \rightarrow 0$. **Decaying oscillations.**

STABILITY CONDITIONS — SUMMARY

$0 < \lambda < 1$ stable, monotone decay	$-1 < \lambda < 0$ stable, alternating decay
$\lambda > 1$ unstable, monotone growth	$\lambda < -1$ unstable, alternating growth

Complex $\lambda = a + bi$: $\rho = \sqrt{a^2 + b^2}$. Stable iff $\rho < 1$. Always oscillatory.
For 2nd-order: need both roots $|\lambda_{1,2}| < 1$.

L'HÔPITAL + TAYLOR REFERENCE

L'Hôpital: if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim = \lim \frac{f'(x)}{g'(x)}$.
Applied to $h(\Delta t) = \frac{2\Delta t}{e^{\Delta t} - 1 + \Delta t}$ as $\Delta t \rightarrow 0$:
Top derivative: 2. Bottom: $e^{\Delta t} + 1 \rightarrow 2$. Answer: $\frac{2}{2} = 1$.
Taylor: $\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots$
Used to show $(1 - \Delta t)^{t/\Delta t} \rightarrow e^{-t}$:
 $\exp\left(\frac{t}{\Delta t} \left(-\Delta t - \frac{(\Delta t)^2}{2} - \dots\right)\right) = \exp(-t - \frac{t\Delta t}{2} - \dots) \rightarrow e^{-t}$

LOG STEPS FORMULA

Solve $r^n = k$ for n : take log both sides: $n \ln r = \ln k \Rightarrow n = \frac{\ln k}{\ln r}$. Always round **up** to next integer.
Pond: $25(0.96)^n = 5 \Rightarrow n = \frac{\ln 0.2}{\ln 0.96} \approx \frac{-1.609}{-0.040} = 39.4 \rightarrow$ **40 weeks**
Drug 90%: $(\frac{1}{3})^n = 0.1 \Rightarrow n = \frac{\ln 0.1}{\ln \frac{1}{3}} = \frac{-2.303}{-0.585} = 8.0 \rightarrow$ **8 days**
Euler 1%: $(0.9)^n < 0.01 \Rightarrow n > \frac{\ln 0.01}{\ln 0.9} \approx \frac{-4.605}{-0.105} = 43.7 \rightarrow$ **44 steps**

KEY ALGEBRA

Quadratic: $\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Modulus: $|a + bi| = \sqrt{a^2 + b^2}$ **Euler identity:** $e^{i\theta} = \cos \theta + i \sin \theta$
Geometric sum: $\sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}$

MARK SCHEME TIPS

- Show all substitution steps — method marks for working not just final answer
- Fixed pts: write value of $f'(x_f)$ explicitly, then state stable/unstable AND alternating/non-alternating
- 2-cycle: write $g'(a) = f'(a) \cdot f'(b) = \dots$ explicitly before concluding
- Euler limit: do all 3 parts separately and show each ($a^n \rightarrow e^t$, $(1 - \Delta t)^n \rightarrow e^{-t}$, $h \rightarrow 1$)
- 2nd-order fixed pt: show linearisation step explicitly (Taylor drop q^2)
- Applied models: state what x_n means physically (marks available for this)
- Complex roots: compute $\rho = |\lambda|$ numerically, then state stable/unstable
- "Describe behaviour": always give stable/unstable + alternating/not + converges to what
- When rounding for "how many weeks": always **round up**, not down

INTEGRATING FACTOR (ODE BACKGROUND)

Form $\frac{dx}{dt} + P(t)x = Q(t)$. Integrating factor: $\mu = e^{\int P dt}$. Multiply both sides: $\frac{d}{dt}(\mu x) = \mu Q$. Integrate both sides, solve for x .
Used to solve ODE directly (alternative to verifying). E.g. $\frac{dx}{dt} + x = 2e^t$: $\mu = e^t$, so $\frac{d}{dt}(e^t x) = 2e^{2t}$, giving $e^t x = e^{2t} + C$, i.e. $x = e^t + Ce^{-t}$. With $x(0) = 0$: $C = -1$, so $x = e^t - e^{-t}$.